Meddi Moumouni

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ISAE-SUPAERO aerospace engineering student specializing in Guidance, Navigation & Control (GNC), Control Systems, and Space Mission Analysis. Bridging control theory with hands-on systems engineering, I focus on the practical modeling, simulation, and implementation of complex orbital missions. My ambition is to contribute to next-generation space programmes as a Mission Analysis or GNC Engineer for space systems.

Education & Training

ISAE-SUPAERO — Toulouse, France
Engineering Diploma (Aerospace) – MSc Equivalent — *Sept. 2023 – July 2027*
- **Major:** GNC, Control Theory & Space Systems Design & Operations.
- **Main Courses:** Space Mechanics, Control Systems, Space Mission Analysis, Systems Engineering, Signal Processing.
- **Academic Project (French Space Command – CDE):** Developing an operational Rendezvous and Proximity Operations (RPO) maneuver planning tool for GEO & LEO satellites.
- **Engagement & Clubs:** OSE Tutoring & Outreach Program, SupaeroMoon Science Team Leader (Space Robotics for ERC), Supaero Space Section (Rocketry for EuRoC).

University of Colorado Boulder — Remote Learning
Aerospace, Aeronautical & Astronautical Engineering – Graduate Coursework — *Mar. 2026 – Present*
- **Ongoing:** Spacecraft Dynamics and Control Specialization (120h Module) by Pr. Hanspeter Schaub.
- **Completed:** Introduction to Systems Engineering Specialization (65h Module) by Pr. William Van Atten.
- **Completed:** Control Systems Analysis: Modeling of Dynamic Systems Course (15h Module) by Pr. Lucy Pao.

Keio University — Tokyo, Japan
Exchange Research Student (Space Robotics) — *Sept. 2025 – Mar. 2026*
- **Lab:** Space Robotics Group (Genya Ishigami's Lab).
- **Research Project:** Developed robotic excavation models for micro-gravity regolith collection on Phobos using Ansys Rocky and MATLAB Simscape.
- **CS Project:** Implemented Reinforcement Learning control for a quadruped robot using PyTorch.

Lycée Pierre de Fermat — Toulouse, France
CPGE MPSI / PSI (Intensive Preparatory Classes for Grandes Écoles, Ranked #1 for PSI major)* — *Sept. 2021 – July 2023*
- **TIPE Engineering Project:** Thermodynamic Analysis of District Heating Networks and Heat Transfer.

Work Experience

Space Systems Engineering Intern — Mar. 2026 – Aug. 2026
CNES (French Space Agency) – Spaceship FR — *Toulouse, France*
- Developed software for a centralized astronaut habitat supervision system.
- Engineered a Proof of Concept integrating IoT sensors and Home Assistant with the AMAIA AI assistant.
- Participated in the ESA Course on ECSS (European Space Standardization) standards in June 2026.

Planetary Science Research Engineer Intern — Apr. 2025 – Sept. 2025
CNRS - IRAP (French Institute of Research in Astrophysics and Planetology) — *Toulouse, France*
- Developed SHAKER, an open-source Python framework (Pandas, NumPy, SQLite) simulating rock dissolution.
- Observed mission operations workflows for NASA Mars rovers Perseverance and Curiosity (SuperCam/ChemCam).

Analog Astronaut & GreenHab Officer — Feb. 2025 – Mar. 2025
The Mars Society – MDRS (Mars Desert Research Station) Crew 311 — *Hanksville, UT, USA*
- Operated and maintained the GreenHab biological life-support module during a 1-month isolated Martian analog simulation.
- Conducted experiments for multiple research centers including CNES, ESA, KTH, UCF, CNRS, IRAP, and TU Delft.
- Managed human-factor ethics protocols; mission research outcomes presented to the French President and ESA.

Space Science Communicator — July 2024 – Aug. 2024
Cité de l'Espace (Science Museum) — *Toulouse, France*
- Conducted outreach on complex aerospace concepts, Artemis/Apollo missions, and ESA astronaut selection.

Personal Projects

Papaya Pathfinder Autonomous Rover | *Robotics, Mechatronics, CAD, GNC* — 2024 – Present
- Constructed a physical mobile rover testbed based on Papaya Pathfinder to develop practical mechatronics skills.

Independent RPG Game Development | *Unreal Engine, C++, Game Design* — Sept. 2025 – Feb. 2026
- Programmed, designed, and published a complete role-playing game (RPG) using Unreal Engine (Feb. 2026).

Skills, Languages & Interests

Space Systems & Engineering: GNC, Control Theory, Space Mechanics, Mission Architecture, Robotics, ECSS Standards, CAD, MATLAB, Ansys Rocky, Patrius, VTS
Software & Programming: Python, C/C++, Java PyTorch, Git, Linux, Docker, SQL, Object-Oriented Programming (OOP), IoT, LLM Integration, PennyLane (Quantum Computing)
Core Competencies & Soft Skills: Team Collaboration, Project Organization, Cross-Cultural Adaptability, Perseverance
Languages: French (Native), English (C1 – TOEFL ITP 610), Italian (B1), Japanese (A2)
Hobbies & Interests: Trail Running, Swimming, Tennis, Bouldering, Debating Club, Painting, Piano, Space Exploration